

Bachelor of Technology — General Engineering - AI and Robotics

Table 1: General Engineering - AI and Robotics — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	34
	Discipline Electives	32
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech General Engineering - AI and Robotics — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical System Around Us	3-0-0-3	3
2	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
3	EE-201	Electromechanics	2.5-0.5-0-3	3
4	IC-253	Programming and Data Structures	2-0-2-3	3
5	ME-100	Reverse Engineering	0-0-2-1	1
6	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
7	AR-521	Control of Robotic Systems	3-0-0-3	3
8	AR-523	Robot Manipulators	2-1-0-3	3
9	EE-302	Control Systems	3-0-2-4	4
10	ME-305	Design of Machine Elements	3-1-0-4	4
11	ME-309	Theory of Machines	3-1-0-4	4
Total				34

Table 3: B. Tech General Engineering - AI and Robotics — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
6		IC-230	Environmental Science	3-0-0-3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language Class I	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
10		HS-342	German I	3-0-0-3	
11	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
12		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech General Engineering - AI and Robotics — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech General Engineering - AI and Robotics — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical System Around Us	3-0-0-3	3
4	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech General Engineering - AI and Robotics — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech General Engineering - AI and Robotics — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	AR-521	Control of Robotic Systems	3-0-0-3	3
3	DC	AR-523	Robot Manipulators	2-1-0-3	3
4	DC	EE-302	Control Systems	3-0-2-4	4
5	DC	ME-305	Design of Machine Elements	3-1-0-4	4
6	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech General Engineering - AI and Robotics — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech General Engineering - AI and Robotics — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech General Engineering - AI and Robotics — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 32 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 11: General Engineering - AI and Robotics — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	AR-511	Autonomous Mobile Robots	3
2	AR-522	Robot Vision	3
3	CS-212	Design of Algorithms	4
4	CS-315	Artificial Intelligence	4
5	EE-203	Network Theory	3
6	EE-260	Signals and Systems	3
7	EE-511	Computer Vision	4
8	GE-303	Modern Powertrains: Electric, Hybrid & IC/Aeroengines	3
9	IC-241	Material Science for Engineers	3
10	ME-210	Fluid Mechanics	3
11	ME-212	Product Manufacturing Technology	3
12	ME-213	Thermodynamics	4

Table 12: General Engineering - AI and Robotics — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	AR-508	Marine Robotics	3

General Engineering - AI and Robotics — Discipline Electives, Even Semesters (4, 6, 8) (continued)

Sl. No	Course Code	Course Name	Credits
2	AR-513	Unmanned Aerial System	3
3	AR-524	Robot Simulators	3
4	AR-525	Reinforcement Learning for Robotics	3
5	AR-526	Multimodal Learning	3
6	CE-251	Hydraulics Engineering	3
7	CS-671	Deep Learning and Applications	4
8	CS-673	Advanced Computer Vision	4
9	CS-683	Generative AI	4
10	DS-314	Probability Theory and Statistics	4
11	EE-211	Analog Circuit Design	4
12	EE-608	Digital Image Processing	4
13	IC-241	Material Science for Engineers	3
<i>Compulsory project components — counted as DE, not selectable</i>			
14	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
15	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4